

Reference Brew — the instructor’s dialed-in cup — FULLY SHOWN

This is the “complete solution” for a non-code course (Guide §6a, step 2). Every student-facing practice file in this module is degraded *from* this one — the paired fill-in card blanks its decisions, the two-cup experiment varies one of its variables. That’s why it exists, and why it’s worth getting exactly right before you build anything else.

For a hands-on course, “complete” is **three things together**: the sheet with every number and a one-line *why*, the thing itself done right and left where students can measure against it, and your own dated result.

1. The sheet — every number, and why it’s that number

Variable	The value	Why this value
Dose	22 g whole bean, weighed	Fills a V60-02 without crowding the bed; big enough that a 1 g scale error doesn’t matter much
Water	352 g, filtered	22 g at 1:16 — the middle of the sweet spot, strong enough to taste structure, not so strong it hides mistakes
Ratio	1:16	The number students leave with. Easy mental math (dose $\times$ 16) and forgiving of grind error
Grind	Medium, table-salt texture	Coarse enough that a beginner’s slow pour won’t over-extract; fine enough to finish in time
Water temp	~200 °F / just off the boil (30 s off heat)	Hot enough to extract sweetness, below boiling so it doesn’t scorch and turn bitter

Variable	The value	Why this value
Bloom	45 g water, 45 s	About 2× the dose. Lets CO ₂ escape so the water can actually reach the grounds
Pour	Slow circles, finish pouring by 2:00	Keeps the bed even; pouring too fast channels and leaves dry grounds
Total time	3:00–3:30 , drawdown complete	The grind check: much faster means too coarse, much slower means too fine
Target	Sweet, clean, no bitterness in the finish	What they’re aiming at — “good” has to be tasteable, not theoretical

Rinse the filter first (hot water, discard) — removes the papery taste and warms the dripper. Students who skip it taste cardboard and blame the beans.

2. The thing itself — brewed, and left where students can measure against it

Brew this cup **before class** and leave it on the bench beside the setup, labelled. It is the benchmark the whole module compares to:

- Objective CO1’s evidence is “*judged against the reference brew*” — this is that reference. Without a cup on the bench, “judged against” has nothing to point at.
- Students taste it **first**, before they brew anything, so “good” is a memory in their mouth rather than a description in a handout.
- Leave the **used bed** in the dripper too. A flat, evenly-extracted bed is what theirs should look like; a cratered one is the most common tell that they poured too fast.

If the room can’t support a pre-brewed cup, brew it live as segment 1’s demo — but then it must be the first thing that happens, not something they see after their own attempt.

3. Your own dated result — the proof it reproduces

Date	Bean / roast date	Grind setting	Time	Notes
2026-03-14	Ethiopian washed, roasted 2026-03-06	Baratza Encore 18	3:12	Sweet, clean. The target cup.
2026-03-21	Same bean, roasted 2026-03-18	Encore 18	2:48	Fresher beans bloomed harder; ran fast. Went to 16 to pull it back.

Why record this: it is how you find out your recipe is drifting before a room of students does. A “complete solution” that worked once isn’t complete — complete means *reproducible*, and the dated log is the only evidence of that. It also gives you the honest answer when a student’s cup runs fast: “*mine did too, the week the beans were that fresh.*”

4. What gets degraded from this

Student file	What was removed
<code>live_session/paired_brew_FILLIN.md</code>	The decisions — ratio, grind, temp, bloom, time. Steps and the <i>why</i> text stay.
<code>guided_practice/two_cup_experiment.md</code>	One variable is deliberately changed so they taste the difference.
<code>live_session/recipe_card_cheatsheet.md</code>	Nothing removed — this sheet, condensed to what fits in a pocket.

Build this file first and those three fall out of it. Build them first and you’ll find they disagree with each other.